

Day 5 Networking Testing Tools

1. Use the ping command to test connectivity from your computer to www.youtube.com and note the average response time.

Ping Test to www.youtube.com

I opened **Command Prompt** and used the following command:

```
ping www.youtube.com
```

The test produced the following results:

- **Packets Sent:** 4
- **Packets Received:** 3
- **Packet Loss:** 25%
- **Minimum Response Time:** 44 ms
- **Maximum Response Time:** 58 ms
- **Average Response Time:** 48 ms

The average response time when pinging www.youtube.com was **48 ms**, with **25% packet loss**.

2. Run the traceroute (tracert on Windows) command to www.flipkart.com and list all the intermediate hops shown in the output.

```
tracert www.flipkart.com
```

Tracing route to e348826.dsca.akamaiedge.net [23.223.245.49]

over a maximum of 30 hops:

```
 1  5 ms   1 ms   1 ms  Unit [192.168.1.1]
 2  33 ms  21 ms  59 ms  10.240.170.222
 3  *      *      *      Request timed out.
 4 116 ms  57 ms  59 ms  182.78.248.109
 5 157 ms 169 ms  95 ms  116.119.104.89
 6  63 ms  65 ms  63 ms  ae4.r03.border01.ccu01.fab.netarch.akamai.com [2.16.180.241]
 7  *      *      108 ms lo1.r03.stem01.ccu01.fab.netarch.akamai.com [2.16.180.26]
 8  *      107 ms *      lo1.r04.spine03.ccu01.fab.netarch.akamai.com [2.16.180.19]
```

```
9  107 ms  71 ms  65 ms lo1.r03.leaf01.ccu01.fab.netarch.akamai.com [2.16.180.6]
10  *      *      108 ms vlan100.r06.tor01.ccu01.fab.netarch.akamai.com [23.216.135.33]
11  73 ms  65 ms  63 ms a23-223-245-49.deploy.static.akamaitechnologies.com
[23.223.245.49]
```

Trace complete.

3. Use the nslookup tool to find the IP address of www.zomato.com and record the result.

```
nslookup www.zomato.com
```

Server: UnKnown

Address: fe80::26de:8aff:fe79:8d11

Non-authoritative answer:

Name: e70929.dsca.akamaiedge.net

Addresses: 2600:140f:201:12::213:d4c8

2600:140f:201:12::213:d4d2

2600:140f:201:12::213:d4d6

2600:140f:201:12::213:d4cc

2600:140f:201:12::213:d4d5

23.223.245.38

23.223.245.49

Aliases: www.zomato.com

zomato.edgekey.net

4. Test connectivity to another device on your local network (such as your friend's laptop or your phone) using the ping command and describe what happens if the device is offline.

When the device was **online**, I received replies successfully:

- **Packets Sent:** 4
- **Packets Received:** 4
- **Packet Loss:** 0%

- **Average Response Time: 330 ms**

when the device was **offline/disconnected**, I ran the same command again. It showed “**Request timed out**” for all four packets:

- **Packets Sent: 4**
- **Packets Received: 0**
- **Packet Loss: 100%**

```
C:\Users\Harsh>ping 192.168.1.92

Pinging 192.168.1.92 with 32 bytes of data:
Reply from 192.168.1.92: bytes=32 time=817ms TTL=64
Reply from 192.168.1.92: bytes=32 time=61ms TTL=64
Reply from 192.168.1.92: bytes=32 time=316ms TTL=64
Reply from 192.168.1.92: bytes=32 time=128ms TTL=64

Ping statistics for 192.168.1.92:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 61ms, Maximum = 817ms, Average = 330ms

C:\Users\Harsh>ping 192.168.1.92

Pinging 192.168.1.92 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.92:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Users\Harsh>
```

5. Create a short report (5-7 lines) summarizing which tool (ping, traceroute, nslookup) is most useful for diagnosing each of these problems: (a) Website not loading, (b) Slow connection, (c) Suspecting DNS issues.

Ping is most useful for checking whether a website or device is reachable and measuring response time; my ping www.youtube.com test showed an **average of 48 ms** with 25% packet loss.

For a **website not loading**, I would use **ping** first to check basic connectivity to the website.

Traceroute (tracert) is useful for finding where a connection becomes slow or fails along the path to a website, such as when tracing the route to www.youtube.com.

For a **slow connection**, traceroute can help identify a particular network hop causing high delays.

nslookup is most useful when **DNS problems are suspected**, because it checks whether a domain name such as `www.youtube.com` can be resolved to an IP address.

Therefore, **ping checks connectivity and delay, traceroute checks the network path, and nslookup checks DNS resolution.**